

Rudra Gupta

☎ 9075569787 — ✉ kumarguptarudra15@gmail.com — 🔗 linkedin.com/in/rudra-kumar-gupta
🐙 github.com/Rudra-Gupta15 — 🌐 rudra-gupta.vercel.app — 🏠 kaggle.com/rudrakumargupta

Education

Yeshwantrao Chavan College of Engineering, Nagpur
B.Tech — Electronics Engineering | Honor: VLSI Design

CGPA: 7.77

Skills

Languages: Python, JavaScript, C/C++, HTML, CSS, SQL
Technologies: Machine Learning, Deep Learning, Generative AI, LLMs, Agentic AI, Computer Vision, OOP
Frameworks: Flask, FastAPI, React, Vite, Node.js
Tools: VS Code, Google Colab, Anaconda, Ollama, MySQL, Git, GitHub, Postman, ESP32, MQTT
Libraries: OpenCV, TensorFlow/Keras, Scikit-learn, face-api.js, Tkinter, YOLOv8, Leaflet.js

Experience

Junior AI/ML Engineer — Prevoyance IT Solutions Pvt. Ltd. Jan 2026 – Present

- Developed real-time computer vision systems for **sign language recognition** and **violence detection in surveillance footage** using CNN, YOLOv8, and OpenCV, achieving **90% live inference accuracy**.
- Built end-to-end surveillance pipelines including dataset preprocessing, model training, hyperparameter tuning, and a Tkinter-based GUI for real-time monitoring.
- Delivered **20+ projects** across machine learning, full-stack web development, and ESP32/Blynk IoT solutions.

Projects

Smart Waste Management System – Edge AI Monitoring Platform 2026

- Trained a YOLOv8 model to detect and sort waste types from camera footage with 87% accuracy, running fast (28ms/frame) on a low-power edge device (Jetson Orin Nano)
- Connected AI detection with IoT sensors to automate 5 real tasks: confirming pickups, spotting litter, and auto-alerting maintenance issues within 2 seconds – all shown on a live map dashboard.

AI Test Case Generator — Agentic QA Pipeline 2026

- Multi-stage LLM pipeline (FastAPI + React/Vite + local Ollama) converts BRDs, FSDs, and Figma designs into complete QA test suites — isolates reasoning steps per stage to avoid context overload and hallucination.
- Plans test coverage before generation (positive/negative/edge cases per feature); configurable modes tune output between deterministic and exploratory testing — zero manual test-writing, fully offline.

InfraPulse — IT Compliance & Discovery Portal 2026

- Agentless network discovery and compliance portal (FastAPI + JS + SQLite) that scans LAN subnets (CIDR/custom ranges), fingerprints open ports (RDP, SMB, SSH), and identifies live devices.
- Cross-platform audit engine (PowerShell/Bash) capturing 10+ hardware specs, software inventory, login history, battery health, and backup status — featuring real-time dashboards and PDF report exports.

Research Publication

IEEE ICETEMS 2026 — Paper Presentation Mar 2026

- Presented “*Sign Language Translator using ML Algorithms on RGB Color Space*” at the **3rd International Conference on Emerging Trends in Engineering and Medical Sciences**, IEEE Maharashtra Section, YCCE.
- Demonstrated RGB color space superiority (90% accuracy) over Greyscale (~40%) and HSV for real-time ASL recognition using CNN + YOLOv8.

Certifications

- Claude 101** — Anthropic, 2026 — **AI Appreciate & AI Aware** — Intel AI For All × Digital India / CBSE, Feb 2026
- Generative AI for Beginners** — Simplilearn SkillUp, Apr 2026

Achievements & Activities

- Kaggle Datasets Expert** — Rank **220 / 11,018** worldwide (Top 2.0%); 3 Silver medals; highest rank **217**.
- Kaggle Code Expert** — Rank **2,336 / 61,392** worldwide (Top 3.8%); highest rank **2,271**.
- Chrome Web Store Publisher** — 3 live extensions, 500+ total installs: *BalanceTab* (200+ installs, 5.0 rating), *TimeMark* (150+ installs, YouTube timestamps), *Growth Tracker* (150+ installs, behavioural tracker + site blocker).
- Event Head, RoboRace** — Led RoboRace at ICON (Annual College Fest), YCCE; managed logistics and participant coordination.
- Nagpur District Level Chess Player; represented at State Level (Class 8).